GraphMind AI | Web-Based AI Math Differentiator

ONE-PAGE PROJECT REPORT • Educational Technology & AI

TOPIC	TECHNOLOGY	AUTHOR
Functions and Graphs · Grade 9	Python · Streamlit · Plotly · OpenAI API	[Your name]

1. Aim and core mathematical task

The app differentiates **one shared learning objective:** investigate a quadratic graph **$f(x) = ax^2 + bx + c$**, identify its vertex, and determine whether the parabola opens upward or downward. Difficulty, context, support and feedback adapt to learner readiness; the core skill stays the same.

2. AI integration and prompt design

The Python backend calls the **OpenAI Responses API** using structured output. The generation prompt requests coefficients, a short contextual question, two progressive hints and a learning tip. Key instructions are: “Use simple A2–B1 English,” “Follow the coefficient limits for the chosen level,” and “Do not reveal the vertex in hints.” After the student submits an answer, a second prompt asks for **1–3 sentences of supportive, level-specific feedback**, using independently checked correctness flags.

3. How differentiation is achieved

LEVEL 1 · FOUNDATION	LEVEL 2 · STANDARD	LEVEL 3 · ADVANCED
$\pm x^2 + c$ Simple shifted graphs. Direct and gradual hints introduce the turning point.	**$ax^2 + bx + c$** A nonzero linear term. Learners apply the vertex formula independently.	**$h(t) = at^2 + bt + c$** Real-world ball height. Interpret maximum height and corresponding time.

4. Interaction, feedback and mathematical reliability

Learners select a level, generate a fresh AI problem, explore a **Plotly interactive graph**, enter the vertex coordinates and opening direction, request hints, receive automatic marking and AI feedback, then reveal a step-by-step solution. To avoid relying on AI arithmetic, Python validates coefficients, calculates vertices using exact rational arithmetic, draws graphs and grades three responses with a stated tolerance.

5. Testing and submission

Offline automated checks cover exact vertex results, level constraints, answer marking and rejection of invalid AI coefficients. A final **live API test with the author’s own key** and a hosted-site test are still required before submission. Deliverables: deployed website URL, a public GitHub repository with README, and this one-page report.

WEBSITE: _____ **GITHUB:** _____